origin . Here the spring midpoint
is fixed. form two fixed-end spring oscillators:
natural length , stiffness . Likewise
form two oscillators, each of natural length and
stiffness .

For , are

displacements from their own equilibria. Then

where

are constants to determine.

where

are constants to determine.

At , relative to the center-of-mass frame,

These are the initial conditions.

Reference circles give four center-frame velocities,
respectively:

Displacements, velocities and initial conditions give

Hence